

Orthogonality cliques over $\mathbb{F}_2$: a radical quotient proof and a coloring separation

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Working note, 28 September 2026

Abstract

We determine the clique number of the graph on all nonzero subspaces of $\mathbb{F}_2^n$, adjacent when they are mutually orthogonal for the standard dot form, answering Nikandish's Problem 4.2. Summing the radicals of a clique and passing to the resulting orthogonal quotient gives the sharp upper bound. We also give explicit coloring certificates in dimension six: the graph on lines has a 14-coloring, whereas the full subspace graph has chromatic number between 15 and 16. Thus the two chromatic numbers already differ in this dimension.

1 The clique number

Let $V = \mathbb{F}_2^n$ with $B(x, y) = \sum_i x_i y_i$. Following Nikandish [1], let $\mathcal{O}_n^*$ have all nonzero subspaces of V as vertices. Distinct U, W are adjacent if $B(U, W) = 0$. Let $N(r)$ denote the number of nonzero subspaces of $\mathbb{F}_2^r$, with $N(0) = 0$.

Theorem 1. *For every integer $n \geq 1$,*

$$\omega(\mathcal{O}_n^*) = \max\{n, N(\lfloor n/2 \rfloor) + (n \bmod 2)\}.$$

Proof. Take a clique $\mathcal{C}$ and set

$$R = \sum_{U \in \mathcal{C}} \text{rad}(U), \quad \text{rad}(U) = U \cap U^\perp, \quad r = \dim R.$$

Each summand is orthogonal to its own member and, by the clique condition, to every other member. Consequently

$$U \subseteq R^\perp, \quad R \subseteq U^\perp, \quad U \cap R = \text{rad}(U) \quad (U \in \mathcal{C}).$$

For the last equality, one inclusion follows from the definition of R ; the other follows from $R \subseteq U^\perp$. Nondegeneracy of B gives $2r \leq n$ and $\dim(R^\perp/R) = n - 2r$.

The form descends to $Q = R^\perp/R$. Write $\bar{U}$ for the image of U in Q . Its restricted form is nondegenerate: if $u \in U$ represents a vector annihilating $\bar{U}$, then $u \in \text{rad}(U) \subseteq R$, so its image is zero. The spaces $\bar{U}$ are pairwise orthogonal. Their nonzero members form an indexed direct sum. Indeed, in a relation $\sum_U x_U = 0$ with $x_U \in \bar{U}$, pairing with arbitrary vectors of one $\bar{W}$ and using its nondegeneracy gives $x_W = 0$. Thus there are at most $n - 2r$ nonzero images. This argument also applies when a restricted or quotient form is alternating.

The members with zero image are precisely those contained in R , and there are at most $N(r)$ of them. Therefore

$$|\mathcal{C}| \leq N(r) + n - 2r. \tag{1}$$

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For $r = 0$ this is n . For every $s \geq 1$, embedding $\mathbb{F}_2^s$ as a hyperplane in $\mathbb{F}_2^{s+1}$ gives

$$N(s+1) \geq N(s) + 2 :$$

an outside line and the whole ambient space are two distinct additional subspaces. Hence $N(s) - 2s$ is nondecreasing for $s \geq 1$. If $r \geq 1$, putting $t = \lfloor n/2 \rfloor$ in (1) gives

$$|\mathcal{C}| \leq N(t) + n - 2t = N(t) + (n \bmod 2).$$

These two cases prove the upper bound.

The n coordinate lines attain the first lower bound. For the second, let $t = \lfloor n/2 \rfloor$ and take

$$T = \text{span}\{e_1 + e_2, e_3 + e_4, \dots, e_{2t-1} + e_{2t}\}.$$

This is a totally isotropic t -space. All its nonzero subspaces are pairwise orthogonal and give a clique of size $N(t)$. When n is odd, $\dim T^\perp = t + 1$, so $T^\perp$ is a distinct additional vertex orthogonal to every subspace of T . This includes $n = 1$, where $T = 0$. The two constructions attain both terms of the maximum. $\square$

Nikandish [1] gives the coordinate and isotropic lower constructions, including the extra perpendicular-space vertex at $n = 7$. The argument above supplies the general upper bound. In particular, it treats overlapping and degenerate clique members without replacing them by lines.

2 A strict coloring separation in dimension six

Let Γ_n be the graph on nonzero vectors of $\mathbb{F}_2^n$, adjacent when their dot product is zero. It identifies with the induced graph on lines in $\mathcal{O}_n^*$. The equalities of chromatic numbers at $n = 4, 5$ in [1] do not continue to $n = 6$.

Proposition 2. *There are explicit certificates for*

$$\chi(\Gamma_6) \leq 14 < 15 = \omega(\mathcal{O}_6^*) \leq \chi(\mathcal{O}_6^*) \leq 16.$$

Proof. Represent a vector by the integer whose six binary digits are its coordinates. The following fourteen sets partition $\{1, \dots, 63\}$. Every two distinct vectors within a row have dot product one, so each row is an independent set in Γ_6 .

Color	Vectors	Color	Vectors
1	2, 14, 50, 62, 63	8	11, 22, 34, 37, 56
2	8, 28, 41, 43, 61	9	4, 5, 52, 54, 60
3	6, 12, 42, 53, 59	10	16, 18, 21, 24
4	19, 38, 46, 55	11	1, 9, 23, 39, 51
5	32, 35, 44, 47, 48	12	15, 30, 40, 49
6	7, 31, 33, 36, 58	13	25, 27, 29, 57
7	13, 17, 20, 26, 45	14	3, 10

This proves $\chi(\Gamma_6) \leq 14$. The totally isotropic space $T = \text{span}\{e_1 + e_2, e_3 + e_4, e_5 + e_6\}$ has seven lines, seven planes and one three-dimensional subspace. These fifteen vertices form the clique already exhibited in [1]; Theorem 1 also gives its optimality.

For the upper bound on the full graph, the accompanying file `coloring-certificate.json` assigns one of sixteen colors to each nonzero subspace, specified by a basis. The independent

checker enumerates the span of each supplied basis and verifies independence and uniqueness. The numbers of distinct subspaces in dimensions 1 through 6 are

$$63, \quad 651, \quad 1395, \quad 651, \quad 63, \quad 1.$$

These agree with the Gaussian subspace counts, so the list contains all 2824 nonzero subspaces. For each pair of distinct subspaces assigned the same color, the checker finds a pair of basis vectors with dot product one. Bilinearity therefore excludes orthogonality of the two subspaces. This verifies a proper 16-coloring of the complete graph. $\square$

The exact values of $\chi(\Gamma_6)$ and $\chi(\mathcal{O}_6^*)$ are not inferred from these colorings. The immediate remaining question for the full graph is whether a 15-coloring exists. A search can assign distinct fixed colors to the fifteen subspaces of T , eliminating color-permutation symmetry, and ask whether the precoloring extends to all remaining subspaces. If no extension exists, a checkable obstruction would establish the value 16; an extension would establish 15.

3 Proof files and reproduction

The complete Lean proof of Theorem 1 is supplied as `NikandishClique.lean`. The public [pinned source](#) is in the trureturing repository at revision

063bb60d1df414c93252023dbc5a5ae659566e58.

Its imports use Lean 4.33.0 and the pinned Mathlib dependencies. The theorem is `D5.S3.Combinatorics.Orthogonality.NikandishClique.result`. The published formal verification records the standard axiom closure `propext`, `Classical.choice`, and `Quot.sound`.

The new colorings are finite certificates checked with exact integer arithmetic; they are not yet Lean formalized. From the accompanying package, run `python3 check_certificate.py`. To regenerate the certificates, first run `python3 coloring_certificate.py`. The checker does not import the search program. AI tools assisted the exposition, code development and computational exploration.

References

- [1] R. Nikandish, *Annihilating-Ideal Graphs and Orthogonality Graphs over $\mathbb{F}_2$* , arXiv:2609.22769v1 (2026).